

The Pretentious Approach to ANT

I

Penon's Formula Revisited

Notes from a lecture by Andrew Granville

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classical proof of the PNT:

$$\Lambda(n) \longrightarrow \psi(n) \xrightarrow{\text{Perron}} -\frac{\zeta'}{\zeta}(s) \xrightarrow{\text{shift contour}} \{p: \zeta(p)=0\} \longrightarrow \text{error term in the PNT}$$

pretentious route keeps the analysis in the half-plane $\text{Re } s > 1$:

Euler product $\longrightarrow$ distance between multiplicative functions $\longrightarrow$ zero-free line

Perron integral on $\text{Re } s > 1 \longrightarrow$ oscillation of $x^s \longrightarrow$ integration by parts

poles of $\frac{\zeta'}{\zeta} \longrightarrow$ Taylor radius on $\text{Re } s > 1 \longrightarrow$ pretentious RH.

classical viewpoint

continue $f(s)$ and shift the contour left

zeros appear as residues of $-\frac{s'}{s}$

the zero-free line follows from a positive trigonometric polynomial

cancellation is encoded by residues and the distribution of zeros

the error term is controlled by the rightmost zero.

pretentious viewpoint

keep the Perron integral on $\text{Re } s > 1$.

zeros appear as possible poles detected by Taylor series.

the same positivity follows from a triangle inequality for Π .

cancellation is extracted directly from the rapid rotation of x^s .

the error term is controlled by derivatives and mean values on $\text{Re } s > 1$.

Lecture topics

Motivation and the scope of the pretentious program

$\Lambda(n)$, $\psi(x)$, and PNT error terms

Perrow's formula and Dirichlet series

Contour shifting, the explicit formula, and zero-free regions

The zero-free line through pretentious distance

Perrow without contour shifting; integration by parts

A pole-cancelling function and the pretentious RM

1. Perron's formula is not (intrinsically) tied to contour shifting.
it is also a tool for exposing the oscillation of x^s on the original
line of absolute convergence.

2. the classical Mertens-de la Vallée Poussin inequality

$$\zeta(6)^2 \left| \zeta(6+it) \right|^4 \left| \zeta(6+2it) \right| \geq 1$$

is a consequence (geometric) of the triangle inequality for
pretentious distance.

3. the line $\operatorname{Re} s = 1$ is zero-free because a hypothetical zero at $1+it$
would force incompatible phase behavior at $1+2it$.

4. Integration by parts changes the Perron problem from a pointwise bound on $A(s)$ to averaged control of $\frac{A'(s)}{A(s)}$, with an $(\log x)^{-1}$ gain.

5. After canceling the pole at $s=1$, RH becomes a statement about Taylor coefficients evaluated entirely in $\text{Re } s > 1$:

$$\frac{Z^{(k)}(s)}{k!} \ll_s 2^k.$$

Prime-counting functions and target error terms

Def 2.1. von Mangoldt function

$$\Lambda(n) = \begin{cases} \log p, & n = p^k, \quad p \text{ prime}, \quad k \geq 1. \\ 0, & \text{otherwise} \end{cases}$$

its summatory function $\psi(x) := \sum_{n \leq x} \Lambda(n)$

$$\text{PNT} \iff \psi(x) \sim x$$

the advantage of ψ over $\pi(x)$ is that its Dirichlet series is a logarithmic derivative:

$$-\frac{\zeta'}{\zeta}(s) = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}, \quad \operatorname{Re} s > 1. \quad (1)$$

$$\begin{aligned} \lceil \quad \zeta(s) &= \prod_p (1 - p^{-s})^{-1} \xrightarrow[\text{in } \operatorname{Re} s > 1]{\text{absolute convergence}} \log \zeta(s) = - \sum_p \log(1 - p^{-s}) = \sum_p \sum_{m \geq 1} \frac{1}{m p^m s}, \\ \frac{\zeta'}{\zeta}(s) &= - \sum_p \sum_{m \geq 1} \frac{\log p}{p^m s} = - \sum_{n \geq 1} \frac{\Lambda(n)}{n^s} \quad \rfloor \end{aligned}$$

3 levels of precision

$$\psi(x) = x + O(x^{3/2} \log^2 x)$$

under RH. (2)

$$\psi(x) = x + O\left(x \exp\left\{-c \frac{(\log x)^{3/5}}{(\log \log x)^{4/5}}\right\}\right)$$

unconditionally (3)

Vinogradov - Korobov form

$$\psi(x) = x + O(x e^{-c\sqrt{\log x}})$$

weaker, but its dependence on a zero-free region is especially transparent

from the classical elementary zero-free region (4)

Perron's formula

the step function as a contour integral

$$c > 0, y > 0 \quad \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{y^s}{s} ds = \begin{cases} 1, & y > 1, \\ 1/2, & y = 1, \\ 0, & 0 < y < 1. \end{cases} \quad (5)$$

$$\boxed{y^s = e^{s \log y}}$$

$y > 1 \implies e^{s \log y}$ decays when the contour is closed to the left.
the contour encloses the simple pole at $s=0$, whose residue is 1.

$0 < y < 1 \implies e^{s \log y}$ decays when the contour is closed to the right.
no pole is enclosed.

the truncated kernel satisfies, for $y \neq 1$,

$$\frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{y^s}{s} ds = \mathbb{1}_{y>1} + O\left(y^c \min\left\{1, \frac{1}{T|\log y|}\right\}\right). \quad (6)$$

$|\log y|^{-1}$ measures the difficulty near the discontinuity $y=1$.

Summing the coefficients of a Dirichlet series

let $A(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$ converge absolutely on $\operatorname{Re} s = c$.

assume for simplicity that $x \notin \mathbb{N}$.

$$\begin{aligned} \mathbb{1}_{n \leq x} &= \mathbb{1}_{x/n > 1} \xrightarrow{(5)} \sum_{n \leq x} a_n = \sum_{n \geq 1} a_n \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{(x/n)^s}{s} ds \\ &= \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \left(\sum_{n \geq 1} \frac{a_n}{n^s} \right) \frac{x^s}{s} ds \\ &= \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} A(s) \frac{x^s}{s} ds \end{aligned} \quad (7)$$

absolute convergence justifies the exchange of summation and integration.

taking $A(s) = -\frac{\zeta'(s)}{\zeta(s)}$.

$$\psi(x) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} -\frac{\zeta'(s)}{\zeta(s)} \frac{x^s}{s} ds + O\left(\frac{x}{T} \log^2(xT) + \log x\right), \quad c > 1. \quad (8)$$

for suitable choice of T .

the exact truncation error depends on how close x is to an integer
this technicality does not affect the lecture's main comparison

the classical contour shift and the explicit formula

Poles of the integrand

starting with (8), the classical method shifts the line of integration to the left.

the pole crossed by $-\frac{\zeta'(s)}{\zeta(s)} \frac{x^s}{s}$ are:

1. $s=1$, arising from the simple pole of ζ ;
2. the zeros ρ of ζ .
3. $s=0$, arising from the factor $1/s$.

$$\text{near } s=1, \quad \zeta(s) = \frac{1}{s-1} + \gamma + O(s-1), \quad -\frac{\zeta'(s)}{\zeta(s)} = \frac{1}{s-1} - \gamma + O(s-1).$$

$$\longrightarrow \operatorname{Res}_{s=1} \left(-\frac{\zeta'(s)}{\zeta(s)} \frac{x^s}{s} \right) = x.$$

if ρ is a zero of multiplicity m , $\zeta(s) = (s-\rho)^m g(s)$, $g(\rho) \neq 0$

$$\longrightarrow \frac{\zeta'(s)}{\zeta(s)} = \frac{m}{s-\rho} + \frac{g'(s)}{g(s)} \longrightarrow \operatorname{Res}_{s=\rho} \left(-\frac{\zeta'(s)}{\zeta(s)} \frac{x^s}{s} \right) = -m \frac{x^\rho}{\rho}, \quad \operatorname{Res}_{s=0} \left(-\frac{\zeta'(s)}{\zeta(s)} \frac{x^s}{s} \right) = -\frac{\zeta'(0)}{\zeta(0)}$$

after counting zeros with multiplicity,

$$\longrightarrow \psi^*(x) = x - \sum_{\rho} \frac{x^\rho}{\rho} - \frac{\zeta'(0)}{\zeta(0)} \quad (9)$$

for estimates it is safer to truncate (with harmless modifications at prime powers)

$$\psi(x) = x - \sum_{\substack{s(\rho)=0 \\ |\operatorname{Im} \rho| \leq T}} \frac{x^\rho}{\rho} + O\left(\frac{x}{T} \log^2(xT) + \log x\right). \quad (10)$$

zero regions control PNT errors

$$B(T) := \{ \rho : \zeta(\rho) = 0, 0 \leq \operatorname{Re} \rho \leq 1, |\operatorname{Im} \rho| \leq T \} \quad , \quad \beta(T) := \max_{\rho \in B(T)} \operatorname{Re} \rho.$$

$$|x^\rho| = |x^{\operatorname{Re} \rho} e^{i(\operatorname{Im} \rho) \log x}| = x^{\operatorname{Re} \rho} \Rightarrow \left| \sum_{\rho \in B(T)} \frac{x^\rho}{\rho} \right| \leq \sum_{\rho \in B(T)} \frac{x^{\operatorname{Re} \rho}}{|\rho|} \leq x^{\beta(T)} \sum_{\rho \in B(T)} \frac{1}{|\rho|} \quad (11)$$

$$\text{zero-counting formula} + \text{partial summation} \Rightarrow \sum_{\rho \in B(T)} \frac{1}{|\rho|} \ll \log^2 T.$$

$$\leadsto |\psi(x) - x| \ll x^{\beta(T)} \log^2 T + \frac{x}{T} \log^2(xT). \quad (12)$$

Under RH. $\text{zf } \beta(T) = \frac{1}{2}$, choose $T = x^{1/2} \leadsto \psi(x) = x + O(x^{1/2} \log^2 x)$.

From the classical zero-free region.

$$\text{Suppose } \beta(T) \leq 1 - \frac{c}{\log T} \leadsto x^{\beta(T)} \leq x \exp\left(-c \frac{\log x}{\log T}\right)$$

$$\text{ignore powers of logarithms, (12)} \Rightarrow |\psi(x) - x| \ll x \exp\left(-c \frac{\log x}{\log T}\right) + x \exp(-\log T).$$

$$\text{balance the 2 exponents, } \frac{\log x}{\log T} = \log T \Rightarrow \log T \asymp \sqrt{\log x}$$

$$\leadsto \psi(x) = x + O(x e^{-c' \sqrt{\log x}}).$$

[this calculation isolates the classical bottleneck: the quality of PNT is governed by how far zeros are kept from the line $\operatorname{Re} s = 1$.]

the zero-free line as a distance inequality

the phase heuristic

$$\text{for } \sigma > 1, \quad \zeta(\sigma + it) = \prod_p (1 - p^{-\sigma - it})^{-1}.$$

if $\zeta(1 + it)$ were zero, then as $\sigma \downarrow 1$, $\log |\zeta(\sigma + it)| \rightarrow -\infty$.

$$\log |\zeta(\sigma + it)| = \sum_p \sum_{m \geq 1} \frac{\cos(mt \log p)}{m p^{m\sigma}}$$

heuristically, $p^{-it} \rightarrow$ negative real axis
squeeze them $\rightarrow +1$.

$\Rightarrow \zeta(1 + 2it)$ should blow up

but $\zeta(s)$ has no pole at $1 + 2it$ when $t \neq 0$.

the distance formalism makes this heuristic precise.

a weighted pretentious distance

Def 5.1. $a_n \geq 0$, $f, g : \mathbb{N} \rightarrow \mathbb{D} : \{z : |z| \leq 1\}$. def

$$D_a(f, g)^2 := \sum_n a_n (1 - \operatorname{Re}(f(n) \overline{g(n)})). \quad (13)$$

for the usual truncated pretentious distance,

$$D_a(f, g; x)^2 := \sum_{p \leq x} \frac{1 - \operatorname{Re}(f(p) \overline{g(p)})}{p}$$

this is not literally a metric on the whole closed unit disk :

if $|f(n)| < 1$, then $D_a(f, f)$ need not vanish.

its decisive metric feature is the triangle inequality

$$D_a(f, w) \leq D_a(f, g) + D_a(g, w) \quad (14)$$

in the unit-modulus case used below, this follows directly from

$$1 - \operatorname{Re}(z\bar{w}) = \frac{1}{2} |z - w|^2, \quad |z| = |w| = 1.$$

and

$$\left(\sum_n \frac{a_n}{2} |f(n) - h(n)|^2 \right)^{1/2} \leq \left(\sum_n \frac{a_n}{2} |f(n) - g(n)|^2 \right)^{1/2} + \left(\sum_n \frac{a_n}{2} |g(n) - h(n)|^2 \right)^{1/2}$$

(Minkowski's inequality in the weighted ℓ^2 -space)

for the zeta function it is natural to put mass on prime powers:

$$\mathcal{D}_b(f, g)^2 := \sum_p \sum_{m \geq 1} \frac{1 - \operatorname{Re}(f(p^m) \overline{g(p^m)})}{m p^{mb}}, \quad b > 1 \quad (15)$$

if $s = b + it$, then

$$\begin{aligned} \log \left| \frac{\zeta(b)}{\zeta(b+it)} \right| &= \operatorname{Re} \log \frac{\zeta(b)}{\zeta(b+it)} \\ &= \sum_p \sum_{m \geq 1} \frac{1 - \cos(mt \log p)}{m p^{mb}} \\ &= \mathcal{D}_b(1, h^{it})^2. \end{aligned} \quad (16)$$

recovering Mertens' inequality

$$L_j := \log |S(b+jit)| \quad , \quad j = 0, 1, 2.$$

$$(16) \Rightarrow ID_b(1, w^{it})^2 = L_0 - L_1. \quad ID_b(1, w^{2it})^2 = L_0 - L_2.$$

a first application of the triangle inequality is

$$ID_b(1, w^{2it}) \leq ID_b(1, w^{it}) + ID_b(w^{it}, w^{2it}) = 2 ID_b(1, w^{it}).$$

↓
invariance under multiplication by a unit-modulus function

$$\xrightarrow{\text{square}} L_0 - L_2 \leq 4(L_0 - L_1) \Leftrightarrow 3L_0 - 4L_1 + L_2 \geq 0$$

$$\xrightarrow{\text{exponentiate}} \frac{|S(b)|^3 |S(b+2it)|}{|S(b+it)|^4} \geq 1. \quad (17)$$

this is close to the desired inequality
but has wrong sign in the middle exponent

to repair it, insert $-n^{it}$ as the intermediate function :

$$\begin{aligned} \mathbb{D}_b(1, n^{2it}) &\leq \mathbb{D}_b(1, n^{-it}) + \mathbb{D}_b(-n^{it}, n^{2it}) \\ &= 2\mathbb{D}_b(1, n^{-it}). \end{aligned}$$

$$\mathbb{D}_b(1, -n^{it})^2 = L_0 + L_1$$

$$\longrightarrow L_0 - L_2 \leq 4(L_0 + L_1) \quad \Leftrightarrow \quad 3L_0 + 4L_1 + L_2 \geq 0$$

$$\xrightarrow{\text{exponentiate}} \zeta(b)^3 \left| \zeta(b+it) \right|^4 \left| \zeta(b+2it) \right| \geq 1. \quad (18)$$

(Mertens inequality)

at the Euler product level, the same positivity is encoded by

$$3 + 4\cos\theta + \cos(2\theta) = 2(1 + \cos\theta)^2 \geq 0.$$

the distance proof reveals the geometric content behind this otherwise fortuitous trigonometric polynomial.

Prop 5.2 (zero-free line)

$$\zeta(1+it) \neq 0 \quad t \neq 0.$$

Pf Suppose $\zeta(1+it) = 0$ with order $m \geq 1$.

$$\text{As } b \downarrow 1, \quad \zeta(b) \asymp (b-1)^{-1}, \quad |\zeta(b+it)| \asymp (b-1)^m.$$

$$t \neq 0 \Rightarrow \zeta \text{ is holomorphic at } 1+2it$$

$$\Rightarrow |\zeta(b+2it)| = O_t(1).$$

$$\Rightarrow \text{LHS of (18) is}$$

$$O_t((b-1)^{-3+4m}) = O_t(b-1) \rightarrow 0,$$

contradicting its lower bound by 1.

□

Perron's formula without crossing the 1-line

a finite Euler product

the classical method begins on $\text{Re } s > 1$ and moves left.

New question: How much can we prove by estimating the same integral where it is?

def

$$\zeta_x(s) := \prod_{p \leq x} (1 - p^{-s})^{-1} = \sum_{\substack{n \geq 1 \\ p|n \Rightarrow p \leq x}} \frac{1}{n^s} \quad (19)$$

its logarithmic derivative contains precisely the prime powers generated by primes $p \leq x$.

after Perron's int off at $n \leq x$, this is enough to recover $\psi(x)$.

choose $c = 1 + \frac{1}{\log x}$, $x^c = ex$.

taking absolute values in the Perron integral gives the upper estimate

$$\begin{aligned} |\psi(x)| &\ll x \max_{|t| \leq T} \left| \frac{\zeta'_x}{\zeta_x}(c+it) \right| \int_{-T}^T \frac{dt}{1+|t|} + |E(x, T)| \\ &\ll x (\log x) (\log T) + |E(x, T)|. \end{aligned} \quad (20)$$

$$\left| \frac{\zeta'_x}{\zeta_x}(c+it) \right| \ll \sum_{\substack{p^m \\ p \leq x}} \frac{\log p}{p^m c} \ll \log x, \quad \int_{-T}^T \frac{dt}{1+|t|} \asymp \log T.$$

although (20) is far from an asymptotic formula, it loses only logarithmic factors over the expected order of magnitude x .

where cancellation comes from

on the vertical line $s = c + it$, $x^s = x^c e^{it \log x}$.

→ x^s completes one rotation when t increases by $\frac{2\pi}{\log x}$.

[indeed, for any s_0 ,

$$\int_{s_0}^{s_0 + \frac{2\pi i}{\log x}} x^s ds = \left[\frac{x^s}{\log x} \right]_{s_0}^{s_0 + \frac{2\pi i}{\log x}}$$

$$= \frac{x^{s_0}}{\log x} (e^{2\pi i} - 1) = 0. \quad] \quad (21)$$

the factor x^s already contains strong cancellation.

the natural way to expose it is to integrate x^s first.

Integration by parts.

$$I = \int A(s) \frac{x^s}{s} ds$$

$$\underline{\underline{dx^s = x^s \log x ds}} \quad \left[\frac{A(s) x^s}{s \log x} \right] - \frac{1}{\log x} \int A'(s) \frac{x^s}{s} ds + \frac{1}{\log x} \int A(s) \frac{x^s}{s^2} ds. \quad (22)$$

the endpoint term is negligible / can be removed by a convenient truncation

the last integral has gained an extra factor $\frac{1}{s}$.

the main term is governed by $A'(s)$, but it also carries the saving $(\log x)^{-1}$.

there is a direct coefficient computation

$$\text{if } A(s) = \sum_{n \geq 1} \frac{a_n}{n^s} \quad \Rightarrow \quad -A'(s) = \sum_{n \geq 1} \frac{a_n \log n}{n^s}$$

Perron's formula turns the 1st integral on the right of (22) into

$$\frac{1}{\log x} \sum_{n \leq x} a_n \log n$$

the complementary part is

$$\sum_{n \leq x} a_n \left(1 - \frac{\log n}{\log x} \right) \quad (23)$$

if $|a_n| \leq 1$

$$\Rightarrow \left| \sum_{n \leq x} a_n \left(1 - \frac{\log n}{\log x} \right) \right| \leq \sum_{n \leq x} \left(1 - \frac{\log n}{\log x} \right)$$

$$\left(\begin{array}{c} \text{Stirling's formula} \\ \log(N!) = N \log N - N + O(\log N) \end{array} \right) = \lfloor x \rfloor - \frac{\log(\lfloor x \rfloor!)}{\log x} << \frac{x}{\log x} \quad (24)$$

→ 1 logarithm is gained before using any subtle cancellation.

if we write $|A'(s)| = |A(s)| \left| \frac{A'}{A}(s) \right|$

$\Rightarrow$ the remaining vertical integral is bounded by

$$\frac{x}{\log x} \max_{\substack{\text{Re } s = c \\ |\text{Im } s| \leq T}} |A(s)| \int_{-T}^T \left| \frac{A'}{A}(c+it) \right| \frac{dt}{1+|t|}. \quad (25)$$

$$\int_{-T}^T \left| \frac{A'}{A}(c+it) \right| \frac{dt}{1+|t|}$$

$$\stackrel{\text{Cauchy-Schwarz}}{\leq} \left(\int_{-T}^T \frac{dt}{1+|t|} \right)^{1/2} \left(\int_{-T}^T \left| \frac{A'}{A}(c+it) \right|^2 \frac{dt}{1+|t|} \right)^{1/2}$$

the lecture does not complete the mean-square estimate at this point.

Integration by parts converts the visible oscillation of x^s into a factor $\frac{1}{\log x}$,

a mean-square control of a logarithmic derivative can improve a $\log x$ loss to roughly $\sqrt{\log x}$.

this method underlies the later proof of Halász's theorem.

The Riemann hypothesis entirely to the right of 1

Cancelling the pole at $s=1$

the Laurent expansions at $s=1$ are

$$\zeta(s) = \frac{1}{s-1} + \gamma + O(s-1) \quad (26)$$

$$\frac{\zeta'(s)}{\zeta(s)} = -\frac{1}{s-1} + \gamma + O(s-1) \quad (27)$$

$$\text{def} \quad Z(s) := \frac{\zeta'(s)}{\zeta(s)} + \zeta(s) - 2\gamma. \quad (28)$$

$$\Rightarrow Z(s) = O(s-1), \quad s \rightarrow 1.$$

$\rightarrow$ the singularity at $s=1$ is removable and $Z(1)=0$.

away from $s=1$, the pole of $\frac{\zeta'}{\zeta}$ are the zeros of ζ , while $\zeta(s)$ is holomorphic there

$$\rightarrow \{ \text{poles of } Z \} = \{ p : \zeta(p) = 0 \}.$$

$$\longrightarrow RH \iff Z(s) \text{ has no poles in } \operatorname{Re} s > \frac{1}{2}. \quad (29)$$

the reverse implication uses the functional-equation symmetry of the nontrivial zeros: a zero with real part $< \frac{1}{2}$ would produce a reflected zero with real part $> \frac{1}{2}$.

Taylor radius and derivative growth

fix s_0 with $\operatorname{Re} s_0 > 1$. write
$$Z(s) = \sum_{k=0}^{\infty} \frac{Z^{(k)}(s_0)}{k!} (s-s_0)^k.$$

by the Cauchy-Hadamard formula, the radius of convergence $R(s_0)$ satisfies

$$\frac{1}{R(s_0)} = \limsup_{k \rightarrow \infty} \left| \frac{Z^{(k)}(s_0)}{k!} \right|^{1/k}$$

$\longrightarrow$ the coefficient bound

$$\left| \frac{Z^{(k)}(s_0)}{k!} \right| \leq C(s_0) 2^k, \quad k \geq 0. \quad (30)$$

$$\Rightarrow R(s_0) \geq 1/2$$

now

$$\bigcup_{\operatorname{Re} s_0 > 1} \left\{ s : |s - s_0| < \frac{1}{2} \right\} = \left\{ s : \operatorname{Re} s > \frac{1}{2} \right\}.$$

$\longrightarrow$ if (130) holds at every s_0 with $\operatorname{Re} s_0 > 1$, $\Rightarrow Z$ has no pole in $\operatorname{Re} s > 1/2$

$\Updownarrow$
RH

Conversely, RH makes Z holomorphic in that half-plane, so Cauchy's estimates on disks of radius just below $1/2$ gives bounds of the form (130).

Def 7.1 (Qualitative pretentious RH)

for every $s = b + it$ with $b > 1$, $\frac{Z^{(k)}(s)}{k!} \ll 2^k$, $k \geq 0$.

this statement refers only to values and derivatives on the right of the 1-line, yet it equivalent to the absence of the zeros of ζ in $\text{Re } s > \frac{1}{2}$.

Remark 7.2. (Quantitative form)

for applications we need uniformity in s .

for every $\varepsilon > 0$, $\left| \left(\frac{\zeta'(s)}{\zeta(s)} + \frac{1}{s-1} \right)^{(k)} \right| \leq C_\varepsilon k! 2^k (1+|t|^\varepsilon)$

in a specified range such as $1 \leq b < 2$ and $0 \leq |t| \leq e^k$.

Under RH we obtain the stronger estimate $\frac{|Z^{(k)}(b+it)|}{k!} \ll 2^k \log(1+|t|)$, $b \geq 1$.

the quantitative statement is what permits a right of 1 proof of PNT error terms.